Computer Networks

Viva Question Bank with Short Model Answers (Units 1 to 5)

Covers all five units from the lecture slides: 138 questions. Answers are kept short and at the level an external examiner usually asks. Questions marked ★ are the most likely ones, so revise those first.

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Unit 1: Introduction to Computer Networks

27 questions | 17 marked as very likely

★ Q1. What is a computer network?

A group of two or more devices (computers, phones, printers) connected by wired or wireless links so they can communicate and share data and resources.

Q2. What are the advantages and applications of a network?

Advantages: resource sharing (files, printers), better communication, flexible access, low cost, internet access. Applications: email, teleconferencing, business and banking, file and directory services.

★ Q3. Differentiate LAN, MAN and WAN.

LAN: small area (lab, campus, office), privately owned, high speed, short delay (about 1-10 km). **MAN**: city-sized, joins several LANs (about 10-100 km). **WAN**: country or world scale, joins LANs and MANs, lower speed, longer delay (beyond 100 km). Example: the Internet.

Q4. Give examples of PAN and WLAN.

PAN: phone connected to Bluetooth earbuds (very short range). WLAN: home or campus Wi-Fi.

★ Q5. What are guided and unguided transmission media?

Guided (wired): twisted pair, coaxial cable, fiber optic. Unguided (wireless): radio waves, microwaves, infrared.

★ Q6. Why are the wires in a twisted pair cable twisted?

Twisting reduces crosstalk and electromagnetic interference. More twists per foot means less noise.

Q7. UTP vs STP?

UTP has no shield: cheap, easy to install, more noise. STP has a metal shield/mesh: higher data rate and better noise protection but costlier.

★ Q8. Compare twisted pair, coaxial and fiber optic cable.

Twisted pair: cheapest, short distance, noise-prone. Coaxial: better shielding and bandwidth than twisted pair, costlier. Fiber: signals travel as light, highest bandwidth, longest distance, lowest attenuation, immune to EMI, but expensive.

Q9. What is the structure of a fiber optic cable?

Core (glass/plastic) carries the light, cladding around it reflects the light back into the core, and a plastic jacket protects it.

★ Q10. Differentiate radio waves, microwaves and infrared.

Radio: omnidirectional, passes through walls (e.g. FM radio). Microwave: unidirectional, line-of-sight, needs aligned antennas (cellular, satellite). Infrared: short range, cannot pass walls, so more secure within a room (TV remote).

Q11. Terrestrial vs satellite microwave?

Terrestrial: tower-to-tower line-of-sight on the ground. Satellite: earth station sends to a satellite, which amplifies and retransmits to another earth station; covers very large areas, but costly to build and has a 12-15 year life.

★ Q12. What is network topology? Name the types.

The arrangement of devices and links in a network. Types: bus, ring, star, mesh, tree, hybrid.

★ Q13. Explain bus, ring, star and mesh topologies with one pro and one con each.

Bus: single backbone cable; cheap, but backbone failure stops everything and collisions occur. **Ring**: devices in a loop, token-based, fewer collisions; one device failure can break the ring and delay grows with nodes. **Star**: all connected to a central switch/hub; easy to manage and one node failure does not matter, but the central device is a single point of failure. **Mesh**: every device linked to others; most reliable, but expensive and complex.

Q14. Which topology gives maximum fault tolerance? What is a hybrid topology?

Mesh. Hybrid is a combination of two or more topologies (e.g. star + bus), giving flexibility and scalability.

★ Q15. Differentiate Internet, Intranet and Extranet.

Internet: public, open to everyone. Intranet: private network inside an organization for employees only. Extranet: limited access for selected outsiders such as business partners.

★ Q16. What are the network edge and network core?

Edge: hosts / end systems (PCs, phones, servers) and edge routers. Core: the mesh of interconnected routers and switches that move data between edges using switching.

Q17. Peer-to-peer vs client-server network?

P2P: no central server, each computer acts as both client and server (BitTorrent). Client-server: clients request, a dedicated always-on server responds (web server, mail server).

★ Q18. Differentiate circuit switching and packet switching.

Circuit: dedicated path set up first (establish, transfer, disconnect); bandwidth reserved, so it can be wasted; suited to voice. Packet: no dedicated path; packets travel independently using store-and-forward; no bandwidth wasted but congestion possible; suited to data.

★ Q19. What is a protocol?

A set of rules that defines the format and order of messages exchanged between network entities, and the actions taken when messages are sent or received.

★ Q20. List the 7 layers of the OSI model with one function each.

7 Application: services to the user (web, mail, file transfer). 6 Presentation: translation, encryption, compression. 5 Session: dialog control, synchronization/checkpoints. 4 Transport: process-to-process delivery, ports, segmentation, flow and error control. 3 Network: logical addressing (IP) and routing of packets. 2 Data link: framing, MAC addressing, error control, hop-to-hop delivery. 1 Physical: transmission of raw bits over the medium.

★ Q21. Differentiate the OSI and TCP/IP models.

OSI has 7 layers, is a generic reference model with separate session and presentation layers, and its network layer supports both connection-oriented and connectionless service. TCP/IP (as taught here) has 5 layers, is protocol-based, has no separate presentation layer, and its network layer is connectionless.

★ Q22. Which devices work at which layer?

Physical: hub, repeater, modem, cables. Data link: switch, bridge. Network: router.

Q23. What are simplex, half-duplex and full-duplex modes?

Simplex: one direction only (keyboard to PC). Half-duplex: both directions but one at a time (walkie-talkie). Full-duplex: both directions at once (telephone).

★ Q24. Explain the four types of delay in a packet-switched network.

$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$. Processing: time to check header and choose the output link. Queuing: waiting in the router buffer, depends on congestion. Transmission = L/R (packet length / link rate). Propagation = d/s (link length / signal speed).

★ Q25. Difference between transmission delay and propagation delay?

Transmission delay is the time to push all bits of the packet onto the link (depends on packet size and link rate). Propagation delay is the time for a bit to travel across the link (depends on distance and medium speed).

Q26. Numerical: a 12000-bit packet on a 1 Mbps link. Transmission delay and max packets per second?

Delay = L/R = 12000 / 10^6 = 12 ms. Max packets per second = 1 / 0.012 = about 83.33.

Q27. What causes packet loss? What is throughput?

Loss happens when a router's queue/buffer is full, and rises with traffic intensity. Throughput is the rate (bits per second) at which data is successfully delivered over a channel.

Unit 2: Application Layer

30 questions | 26 marked as very likely

★ **Q1. What is the application layer? Which protocols belong to it?**

The top layer; the interface between user applications and the network. It defines how applications exchange information but does not transmit data itself. Protocols: HTTP, FTP, SMTP, POP3, IMAP, DNS, DHCP, SNMP, Telnet, SSH.

★ **Q2. Why are application layer protocols important?**

They enable user services (web, email, file transfer), standardize message formats and rules, and give interoperability between different devices and operating systems.

★ **Q3. Explain client-server and P2P architectures.**

Client-server: server is always on, has a fixed IP, clients do not talk to each other directly. P2P: end systems communicate directly and act as both client and server; self-scalable, but management is complex and peers change IPs.

★ **Q4. What is a socket? What identifies a socket address?**

A socket is the software interface (the 'door') between an application process and the transport layer. A socket address is IP address + port number + transport protocol (TCP or UDP).

Q5. What is a process? Client process vs server process?

A process is a program under execution. Client process initiates communication; server process waits to be contacted. Processes on one host use IPC; on different hosts they exchange messages.

★ **Q6. What transport services can an application need?**

Reliable data transfer, throughput, timing (low delay), and security. Loss-tolerant apps (audio/video) accept some loss; bandwidth-sensitive apps need minimum throughput; elastic apps use whatever is available.

★ **Q7. What services do TCP and UDP offer to applications?**

TCP: connection-oriented, reliable ordered delivery, flow control, congestion control; no timing or minimum-throughput guarantee. UDP: connectionless, unreliable, no flow or congestion control, no guarantees, but fast and low overhead.

★ **Q8. Which transport protocol do common applications use?**

Email (SMTP), web (HTTP), file transfer (FTP) and Telnet use TCP. Streaming media may use TCP or UDP. Internet telephony (VoIP) typically uses UDP. DNS and SNMP use UDP.

★ **Q9. Write the port numbers of common protocols.**

HTTP 80, HTTPS 443, FTP 20 (data) and 21 (control), SSH 22, Telnet 23, SMTP 25 (587 for secure submission), DNS 53, DHCP 67/68, POP3 110, IMAP 143, SNMP 161/162.

★ **Q10. What is the Web? What is a URL?**

The Web is a service on the Internet to access documents/objects using a browser. It has three components: browser, web server, web pages. A web page has a base HTML file plus referenced objects (images etc.), each addressed by a URL = host name + path name.

★ **Q11. What is HTTP? Why is it called stateless?**

HyperText Transfer Protocol, the web's application layer protocol (client browser and server exchange request/response messages over TCP, port 80). It is stateless because the server keeps no information about past client requests.

★ **Q12. HTTP vs HTTPS. What is SSL/TLS?**

HTTPS is HTTP running over SSL/TLS, so data is encrypted and the server is authenticated (port 443). SSL/TLS is the security layer providing encryption, authentication and integrity.

★ Q13. Differentiate persistent and non-persistent HTTP.

Non-persistent (HTTP/1.0 default): one TCP connection per object, closed after the response; response time = 2 RTT + file transmission time per object. Persistent (HTTP/1.1 default): connection stays open for many objects, saving repeated handshakes.

Q14. What is RTT?

Round-trip time: the time for a small packet to go from client to server and back.

★ Q15. Describe the HTTP request message format. Name some methods.

Request line (method, URL, HTTP version), header lines (Host, User-Agent, Accept, Connection...), a blank line (CRLF), and an optional body. Methods: GET, POST, HEAD, PUT, DELETE.

★ Q16. Describe the HTTP response message and common status codes.

Status line (version, code, phrase), header lines (Date, Server, Content-Length, Content-Type...), blank line, then the data/entity body. Codes: 200 OK, 301 Moved Permanently, 400 Bad Request, 404 Not Found, 505 HTTP Version Not Supported.

★ Q17. What are cookies? What are their four components?

Small text files stored by the browser so a website can recognize you and remember preferences. Components: cookie header in the HTTP response (Set-Cookie), cookie header in the HTTP request, cookie file on the user's machine, and a back-end database at the website.

★ Q18. What is a web cache (proxy server)? Benefits?

It satisfies requests on behalf of the origin server and keeps copies of recently requested objects. If the object is in cache it is returned directly; otherwise the cache fetches it from the origin server. Benefits: lower response time and less traffic on the institution's access link.

★ Q19. Forward proxy vs reverse proxy vs load balancer?

Forward proxy sits in front of clients, hides their identity and can cache/filter. Reverse proxy sits in front of servers, hides them and can cache, secure and terminate SSL. Load balancer spreads incoming requests across several servers for availability and performance.

★ Q20. Explain FTP. Why does it use two connections?

FTP transfers files using client-server over TCP. A control connection (port 21) carries commands, username and password; a separate data connection (port 20) carries the actual file. The server keeps state about the user (current directory, authentication).

★ Q21. Telnet vs SSH?

Both give remote login. Telnet (port 23) sends everything, including passwords, in plain text. SSH (port 22) encrypts the session using public key and session keys, so it replaced Telnet.

★ Q22. What are the components of the email system? Explain how a mail is sent.

User agents, mail servers (with mailbox and outgoing queue), and SMTP. Alice's user agent sends to her mail server, which queues it and uses SMTP over TCP to deliver it to Bob's mail server, which stores it in Bob's mailbox; Bob reads it with his user agent using POP3 or IMAP.

★ Q23. SMTP vs POP3 vs IMAP?

SMTP (port 25): sends/pushes mail between servers. POP3 (110): downloads mail to the client and normally deletes it from the server. IMAP (143): keeps mail on the server, supports folders, searching and use from many devices. Analogy: SMTP sends the letter, POP3 collects it, IMAP lets you access the same mailbox anywhere.

Q24. Limitations of SMTP? Phases of POP3?

SMTP restricts message body to 7-bit ASCII and uses no intermediate servers; if the receiving server is down the sender retries later. POP3 phases: authorization, transaction, update.

★ Q25. What is SNMP? Explain its architecture.

Application layer protocol to monitor and manage devices (routers, switches, servers) using UDP ports 161/162. Manager (NMS) sends requests; Agent runs on the device; MIB is the database of device information, located using OIDs. Operations: GET, SET, GET-NEXT, TRAP (alert from agent), INFORM (trap with acknowledgement).

★ Q26. What is DHCP? Explain the DORA process.

DHCP automatically assigns IP address and other settings (mask, gateway, DNS) for a lease time. DORA: Discover (client looks for server), Offer (server proposes an IP), Request (client accepts), Acknowledge (server confirms the lease).

★ Q27. What is DNS and how does name resolution work?

DNS translates domain names into IP addresses. A query goes to the local DNS server, then to a root server (gives TLD server), then the TLD server such as .com (gives the authoritative server), and finally the authoritative server returns the IP address.

★ Q28. Iterative vs recursive DNS query?

Iterative: each server replies with the address of the next server to ask and the requester follows the chain itself. Recursive: the contacted server does the whole resolution on behalf of the requester and returns the final answer.

Q29. Why is DNS a distributed hierarchical database and not centralized?

A single server would be a single point of failure, could not handle the traffic volume, would be far from many clients (high delay), and would be hard to keep updated. There are 13 root server addresses at the top of the hierarchy.

★ Q30. Explain socket programming with TCP and UDP.

TCP (SOCK_STREAM): server creates a welcoming socket and waits; client creates a socket and connects (handshake); server accept() returns a new connection socket per client; data is reliable and ordered. UDP (SOCK_DGRAM): no connection; each datagram carries the destination IP and port, and the server replies to the client address it received.

Unit 3: Transport Layer

25 questions | 22 marked as very likely

★ **Q1. What does the transport layer do?**

Provides logical process-to-process communication between applications on different hosts. The sender breaks messages into segments; the receiver reassembles them. Ports identify the application. Protocols: TCP and UDP.

★ **Q2. Transport layer vs network layer?**

Network layer gives logical communication between hosts (IP address). Transport layer gives logical communication between processes (port numbers). Analogy: the postal service is the network layer; the person who hands letters to the right cousin in the house is the transport layer.

★ **Q3. What are multiplexing and demultiplexing?**

Multiplexing (sender): collect data from several sockets and add a transport header. Demultiplexing (receiver): use header fields (ports, IPs) to deliver each segment to the correct socket.

★ **Q4. How does demultiplexing differ in UDP and TCP?**

UDP: uses the destination port (and IP); segments with different source IPs/ports but the same destination port go to the same socket. TCP: a socket is identified by the 4-tuple (source IP, source port, destination IP, destination port), so a web server has a separate socket per client.

★ **Q5. What is UDP? Where is it used?**

User Datagram Protocol: connectionless, best-effort, no handshake, no retransmission. Used by DNS, SNMP, RIP, online gaming, streaming and VoIP (loss-tolerant, delay-sensitive).

★ **Q6. Describe the UDP header.**

Four 16-bit fields: source port, destination port, length, checksum. Total header = 8 bytes.

★ **Q7. If UDP is unreliable, why does it have a checksum? How is it calculated?**

UDP gives no ACK or retransmission, so the checksum is its only way to detect corrupted data. Sender: treat the segment as 16-bit words, add them with 1's complement (carry wraps around), complement the result. Receiver: add all words including the checksum; if all bits are 1 there is no error.

★ **Q8. Differentiate TCP and UDP.**

TCP: connection-oriented, reliable, ordered, flow and congestion control, 20-byte minimum header, slower; used for web, email, FTP. UDP: connectionless, unreliable, no control mechanisms, 8-byte header, fast; used for streaming, DNS, gaming.

★ **Q9. What is reliable data transfer (rdt)? Name the versions.**

Guarantee that data arrives without loss, duplication or errors and in order. rdt 1.0: perfect channel. rdt 2.0: bit errors, using checksum, ACK/NAK, retransmission (stop-and-wait). rdt 2.1: sequence numbers (0/1) to handle duplicates and corrupted ACK/NAK. rdt 2.2: ACK-only, a duplicate ACK acts as NAK. rdt 3.0: lossy channel, adds timer and retransmission.

★ **Q10. Explain rdt 3.0 (alternating-bit protocol).**

Sender sends packet with sequence number 0 or 1 and starts a timer. If the correct ACK arrives it sends the next packet with the other number; if the timer expires (data or ACK lost) it retransmits. The receiver detects duplicates using the sequence number and re-ACKs.

★ **Q11. Why is stop-and-wait inefficient?**

The sender sends one packet and waits for its ACK, so it is idle most of the time. Example: 1000-byte packet on 10 Mbps takes 0.8 ms to send but RTT is 100 ms, so utilization is under 1%.

★ Q12. What is pipelining and the sliding window?

Sender transmits multiple packets without waiting for ACKs. The window is the range of sequence numbers the sender may send before an ACK is received. Two forms: Go-Back-N and Selective Repeat.

★ Q13. Explain Go-Back-N.

Sender may have up to N unACKed packets and keeps one timer for the oldest. Receiver sends only cumulative ACKs and discards out-of-order packets. On timeout the sender retransmits that packet and all following ones.

★ Q14. Explain Selective Repeat.

Sender keeps a timer per packet and retransmits only the lost one. Receiver ACKs each packet individually and buffers out-of-order packets, delivering them to the upper layer in order.

★ Q15. Go-Back-N vs Selective Repeat?

GBN: cumulative ACK, receiver has no buffering, whole window retransmitted, simpler but wasteful. SR: individual ACKs, receiver buffers, only lost packets retransmitted, more efficient but more complex.

★ Q16. What are the main features of TCP?

Connection-oriented, reliable delivery, ordered data transfer, error checking with checksum, retransmission, flow control and congestion control.

★ Q17. Explain the TCP three-way handshake. Why is it needed?

1) Client sends SYN. 2) Server replies SYN-ACK. 3) Client sends ACK, then data flows. It confirms both sides are ready and synchronizes initial sequence numbers. Closing uses FIN and ACK segments.

★ Q18. Describe the TCP segment header.

Source port, destination port, sequence number (32 bit), acknowledgement number (32 bit), header length, flags (URG, ACK, PSH, RST, SYN, FIN), receive window, checksum, urgent pointer, options. Header size: minimum 20 bytes, maximum 60 bytes (options up to 40 bytes).

Q19. What do the sequence and acknowledgement numbers mean? What do the flags do?

Sequence number = byte-stream number of the first byte in the segment. Acknowledgement number = next byte the host expects. SYN: set up connection. ACK: ack field valid. FIN: no more data. RST: reset. PSH: push data to the application. URG: urgent data.

★ Q20. Differentiate flow control and congestion control.

Flow control protects the receiver from being overwhelmed; the receiver advertises its window (rwnd). Congestion control protects the network when too many senders overload routers.

★ Q21. What causes congestion?

Too many sources sending too fast for routers with finite buffers. Packets get dropped, timeouts trigger retransmissions and duplicates, which load the network even more.

Q22. What are the approaches to congestion control?

End-to-end: no network feedback, congestion inferred from loss and delay (used by TCP). Network-assisted: routers give feedback, such as a congestion bit (ECN).

★ Q23. Explain TCP slow start.

The congestion window (cwnd) starts small and is increased as ACKs come back (roughly doubling every RTT), until the network's capacity, a loss, or the receiver window limit is reached. The sender uses the smaller of cwnd and rwnd.

Q24. What is the leaky bucket algorithm?

A traffic shaping method: data enters a bucket and leaks out at a constant rate, so bursty traffic becomes smooth. If the bucket is full, excess data is discarded.

★ Q25. What is RPC?

Remote Procedure Call lets a program call a procedure on another computer as if it were local. It is not a transport protocol; it uses TCP (e.g. gRPC) or UDP to carry the request and the reply.

Unit 4: Network Layer

31 questions | 24 marked as very likely

★ **Q1. What are the main functions of the network layer?**

Logical addressing (IP), routing (choosing the path), packet forwarding (moving a packet from a router's input to the correct output), and encapsulating segments into datagrams.

★ **Q2. Forwarding vs routing?**

Forwarding is the local action in a router: look at the packet header and the forwarding table and send it to an output link. Routing is the network-wide process of determining the end-to-end path using routing algorithms.

Q3. What services can a network layer offer?

Guaranteed delivery, delivery with bounded delay, in-order delivery, guaranteed minimum bandwidth, guaranteed maximum jitter, security. The Internet provides best-effort service.

★ **Q4. Virtual circuit vs datagram network?**

Virtual circuit: connection setup, data transfer, teardown; packets carry a VC number and follow the same path; routers keep state; router failure kills its circuits. Datagram: no setup, each packet carries the full destination address and is routed independently; no per-connection state (the Internet).

★ **Q5. Switching vs routing?**

Switching works at Layer 2 using MAC addresses inside one LAN. Routing works at Layer 3 using IP addresses and routing tables to connect different networks.

★ **Q6. What is a router?**

A Layer 3 device that forwards packets between networks using the packet header and its forwarding table; it has multiple interfaces, each with an IP address.

★ **Q7. What are the requirements of routing?**

A routing algorithm (finds the best path), a routing table (stores paths), and a metric (hop count, bandwidth, delay) to compare paths.

★ **Q8. Static, dynamic and default routing?**

Static: administrator enters routes manually; no CPU overhead and secure, but not scalable. Dynamic: routers learn and update routes automatically (RIP, OSPF); adapts to changes but uses CPU and bandwidth. Default: all packets without a specific match go to one next hop; used with a single exit point.

★ **Q9. Describe the IPv4 datagram format.**

Version, header length, type of service, total length, identification, flags, fragment offset, TTL, protocol, header checksum, source IP, destination IP, options, data. Header is 20 to 60 bytes; total datagram up to 65,535 bytes.

★ **Q10. What is TTL?**

Time To Live: an 8-bit hop limit reduced by 1 at every router. When it reaches 0 the packet is discarded (ICMP Time Exceeded sent). It prevents packets looping forever.

Q11. How does IP fragmentation work?

A large datagram is split to fit a link's MTU. Identification marks fragments of the same packet, the MF flag says more fragments follow, DF forbids fragmenting, and the fragment offset gives position in units of 8 bytes.

★ **Q12. What is an IP address? Explain classful addressing.**

A 32-bit address in four octets, split into network ID and host ID. Class A: 0.0.0.0-127.255.255.255, mask 255.0.0.0 (/8). Class B: 128-191, mask 255.255.0.0 (/16). Class C: 192-223, mask 255.255.255.0 (/24). Class D: 224-239, multicast. Class E: 240-255, reserved.

★ Q13. How many hosts per network in classes A, B and C?

Usable hosts = 2^h - 2 (network and broadcast addresses excluded). A: 2^{24} - 2 = 16,777,214. B: 2^{16} - 2 = 65,534. C: 2^8 - 2 = 254.

Q14. What are special IP addresses?

Network address (all host bits 0), broadcast address (all host bits 1), 0.0.0.0 (default/unspecified), 127.0.0.1 (loopback). Private ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.

★ Q15. What is subnetting? How do you count subnets and hosts?

Splitting a network into smaller ones by borrowing host bits. Subnets = 2^N (N = borrowed bits); hosts per subnet = 2^H - 2 (H = remaining host bits). Example 192.168.1.0/26: 2 bits borrowed so 4 subnets, mask 255.255.255.192, block size 64, subnets at 0, 64, 128, 192, and 62 usable hosts each.

★ Q16. What is CIDR?

Classless Inter-Domain Routing: slash notation for the subnet mask (e.g. /27 means 27 network bits). It removes fixed classes, reduces routing table size and wastes fewer addresses.

★ Q17. Numerical: find the network, broadcast and usable range for 205.16.37.39/28.

Block size is 16. Network = 205.16.37.32 (IP AND mask), broadcast = 205.16.37.47. First host .33, last host .46. 16 addresses, 14 usable.

Q18. Which subnet masks are valid?

Valid masks have continuous 1s followed by continuous 0s. 255.255.255.240 and 255.255.0.0 are valid; 255.0.255.0 and 255.230.255.0 are invalid.

★ Q19. What is NAT? Why is it used?

Network Address Translation converts private IPs to a public IP (and back) at the router. It saves public IPv4 addresses and gives basic security by hiding internal hosts. A NAT table maps (public IP, port) to (private IP, port); using ports for this is called PAT.

Q20. Explain NAT terminology.

Inside local: private address of an inside host. Inside global: its public address after translation. Outside local / outside global: destination host address before / after translation.

★ Q21. What is ICMP? Where is it used?

Internet Control Message Protocol reports errors and tests the network. Messages: destination unreachable, time exceeded, parameter problem, redirect, echo request/reply. Ping uses echo; traceroute uses TTL-exceeded.

★ Q22. IPv4 vs IPv6?

IPv4: 32-bit, header checksum, options, broadcast, ARP, fragmentation by routers and sender, needs DHCP or manual configuration. IPv6: 128-bit, no header checksum, extension headers, no broadcast, Neighbor Discovery instead of ARP, flow label, fragmentation only by sender, auto-configuration.

★ Q23. Link state vs distance vector routing?

Link state: every router learns the whole topology (flooding), runs Dijkstra, sends triggered/incremental updates, no loops; OSPF. Distance vector: router knows only distances via neighbours, uses Bellman-Ford, sends full table periodically to neighbours, prone to loops; RIP.

★ Q24. Explain Dijkstra's algorithm briefly.

Start at source u with N' = {u} and set D(v) = link cost for neighbours, infinity for others. Repeatedly pick the node w outside N' with minimum D(w), add it to N', and update D(v) = min(D(v), D(w) + c(w,v)) for its neighbours, until all nodes are in N'.

★ Q25. State the Bellman-Ford equation for distance vector.

$d_x(y)$ = min over neighbours v of { $c(x,v) + d_v(y)$ }: cost to a neighbour plus that neighbour's cost to the destination, minimized over all neighbours.

★ Q26. What is the count-to-infinity problem? How is it reduced?

In distance vector routing, when a link fails, routers keep updating each other with slowly increasing costs ('bad news travels slowly'). Solutions: split horizon, poison reverse, and defining a maximum (RIP uses 16 as infinity).

Q27. What is hierarchical routing?

Routers are grouped into regions (autonomous systems). Each router knows full details only for its own region, which shrinks routing tables and update traffic.

★ Q28. Explain broadcast and multicast routing.

Broadcast (to all): flooding (needs hop count to stop loops), multi-destination addressing, or spanning tree (no duplicates). Uses: DHCP discovery, ARP request. Multicast (to a group): group address (class D), a spanning tree is pruned to members only; used in IPTV, video conferencing.

★ Q29. Compare RIP, OSPF and BGP.

RIP: intra-domain, distance vector, Bellman-Ford, max 16 hops, small networks. OSPF: intra-domain, link state, Dijkstra, large networks. BGP: inter-domain (between autonomous systems), path vector, very large networks.

★ Q30. What are firewall, VPN and IPsec?

Firewall: filters incoming/outgoing traffic by rules. VPN: encrypted private connection over the Internet. IPsec: protocol suite securing IP packets with encryption, authentication and integrity, e.g. site-to-site VPN.

Q31. What is a VPC (Virtual Private Cloud)?

A logically isolated private network inside a public cloud where you control the IP range, subnets, route tables and gateways.

Unit 5: Link Layer and Local Area Networks

21 questions | 16 marked as very likely

★ **Q1. What are the services of the data link layer?**

Framing, link access using MAC addresses, reliable delivery (ACK and retransmission), flow control between adjacent nodes, and error detection and correction.

★ **Q2. What are a node, a link and a frame?**

Hosts and routers are nodes; the channels connecting adjacent nodes are links (wired or wireless). The link layer encapsulates a datagram into a frame (header + trailer) and moves it to the adjacent node.

★ **Q3. What is framing? Explain bit stuffing.**

Framing marks the start and end of frames. Bit stuffing uses the flag 01111110; after every five consecutive 1s in the data the sender inserts a 0, and the receiver removes it. Example: data 01111110 is sent as 011111010.

★ **Q4. Explain byte (character) stuffing.**

A flag byte marks frame boundaries. If the flag appears in the data, the sender inserts an escape byte (ESC) before it, and the receiver removes the ESC before passing data up.

★ **Q5. Name the error detection techniques.**

Parity check, checksum, and cyclic redundancy check (CRC).

★ **Q6. Explain parity check. What is its limitation?**

One extra bit makes the number of 1s even (even parity) or odd (odd parity). It detects any single-bit error, but cannot reliably detect two or more bit errors.

Q7. How does the checksum method work here?

Data is split into k segments of m bits. Sender adds them using 1's complement and sends the complement of the sum as checksum. Receiver adds all segments plus the checksum; result of all 1s (complement zero) means accept.

★ **Q8. Explain CRC with an example.**

Append (k-1) zeros to the data (k = bits of the generator), do modulo-2 (XOR) division, and append the remainder to the data. Receiver divides again: remainder 0 means no error. Example: data 100100, generator 1101 (x^3+x^2+1) gives remainder 001, so the sent frame is 100100001.

Q9. Why is CRC considered powerful?

It is based on binary division by a generator polynomial, is easy to implement in hardware, and detects burst errors and far more errors than parity or checksum.

★ **Q10. What is a multiple access protocol? Name the three categories.**

Rules that control how many nodes share one broadcast channel. Categories: channel partitioning (TDMA, FDMA, CDMA), random access (ALOHA, CSMA, CSMA/CD, CSMA/CA) and taking turns (polling, token passing).

Q11. Point-to-point link vs broadcast link?

Point-to-point has one sender and one receiver. Broadcast link has many senders/receivers on one shared channel, so every node receives each frame and collisions can occur.

★ **Q12. Explain TDMA, FDMA and CDMA. What is the drawback of TDMA?**

TDMA: time divided into frames and slots, one per node. FDMA: each node gets a fixed frequency band. CDMA: each node has a unique code so all can transmit at once (used in cellular). TDMA/FDMA drawback: a node gets only R/N rate and wastes its slot even if others are idle.

★ Q13. Differentiate pure ALOHA and slotted ALOHA.

Pure ALOHA: send any time; collision on overlap; vulnerable time 2T; $S = G\, e^{-2G}$; max throughput about 18.4% at G = 0.5. Slotted ALOHA: time divided into slots, send only at slot start; vulnerable time T; $S = G\, e^{-G}$; max throughput about 36.8% at G = 1.

★ Q14. Explain CSMA/CD.

Carrier Sense Multiple Access with Collision Detection, used in wired Ethernet: listen to the channel, transmit if idle, keep monitoring; if a collision is detected, stop immediately, wait a random backoff time and retry.

★ Q15. Why is CSMA/CA used in Wi-Fi? What are RTS/CTS?

A wireless station cannot reliably hear a collision while transmitting, and hidden nodes exist, so Wi-Fi avoids collisions instead. A sender first sends RTS (Request to Send); the receiver replies CTS (Clear to Send), which makes other nodes wait. This solves the hidden node problem.

Q16. What are polling and token passing?

Polling: a master node asks each node in turn to transmit; no collisions, but master failure stops the network. Token passing: a small token frame circulates in a fixed order and only the holder can send; a lost token or failed node disrupts the channel.

★ Q17. What is Ethernet? Name its types.

The most widely used LAN technology (IEEE 802.3, CSMA/CD). Types: Standard/Switched Ethernet 10 Mbps, Fast Ethernet 100 Mbps (same frame format), Gigabit Ethernet (mainly fiber), 10-Gigabit Ethernet.

★ Q18. MAC address vs IP address?

MAC: 48-bit physical address burned into the NIC, used at the link layer for hop-to-hop delivery within a LAN. IP: logical address at the network layer used for end-to-end delivery across networks.

★ Q19. What is ARP? How does it work?

Address Resolution Protocol maps an IP address to a MAC address on a LAN. The host checks its ARP cache; if no entry, it broadcasts an ARP request ('who has this IP?'); the owner replies with its MAC address; the cache is updated and the frame is sent.

★ Q20. Hub vs switch vs router?

Hub (Layer 1): repeats bits to all ports. Switch (Layer 2): learns MAC addresses and forwards frames only to the right port. Router (Layer 3): connects different networks using IP addresses.

Q21. What is a VLAN?

A virtual LAN logically groups devices into separate broadcast domains on the same physical switch, regardless of physical location, improving security and reducing broadcast traffic.

Bonus: Cross-chapter Questions

4 questions | 4 marked as very likely

★ **Q1. What happens when you type a URL in the browser and press Enter?**

DHCP has already given your IP, mask, gateway and DNS server. 1) DNS resolves the domain to an IP address. 2) TCP three-way handshake with the web server (port 80/443). 3) Browser sends an HTTP GET request. 4) Packets are routed hop by hop using IP and forwarding tables (NAT at the edge router); on each LAN, ARP finds the next hop's MAC address and the frame is sent. 5) Server sends the HTTP response; the browser renders the page.

★ **Q2. Explain encapsulation across the layers.**

Application data becomes a segment (transport header with ports), then a datagram/packet (IP header), then a frame (link header and trailer with MAC addresses), then bits on the wire. Each layer adds its own header and the receiver removes them in reverse.

★ **Q3. Give the PDU name and address type at each layer.**

Application: message. Transport: segment (port). Network: packet/datagram (IP address). Data link: frame (MAC address). Physical: bits.

★ **Q4. Which layers use which error-detection method?**

Link layer: CRC (frame trailer). Network layer: IPv4 header checksum. Transport layer: TCP/UDP checksum.

Quick Reference

Protocol	Port
HTTP	80
HTTPS	443
FTP	20 data, 21 control
SSH	22
Telnet	23
SMTP	25 (587 secure)
DNS	53
DHCP	67 / 68
POP3	110
IMAP	143
SNMP	161 / 162

Item	Formula / value
Transmission delay	L / R
Propagation delay	d / s
Nodal delay	dproc + dqueue + dtrans + dprop
Non-persistent HTTP time	2 RTT + file time (per object)
Subnets / hosts	2^N subnets, 2^H - 2 hosts
Block size	256 - subnet mask octet
Pure ALOHA	S = G e^(-2G), max 18.4%, vulnerable 2T
Slotted ALOHA	S = G e^(-G), max 36.8%, vulnerable T
UDP header	8 bytes
TCP header	20 to 60 bytes
IPv4 header	20 to 60 bytes, total up to 65,535 B
CRC zeros appended	k - 1 (k = generator bits)

Layer cheat line (top to bottom)

Application (HTTP, FTP, SMTP, DNS, DHCP, SNMP) - Presentation - Session - Transport (TCP, UDP: segment, port) - Network (IP, ICMP, routing: packet, IP address) - Data link (Ethernet, ARP, CRC: frame, MAC address) - Physical (bits).